Sieve of Eratosthenes (1–100)

Mathematical description. Fix $n = 100$. Let $S = \{2, 3, \ldots, n\}$. For each integer p defined as the smallest element of S not yet processed, declare p *prime* and remove from S every integer of the form kp with $k \geq 2$. Continue this procedure while $p^2 \leq n$. The elements of S that remain after this process are precisely the prime numbers $\leq n$.

Algorithm (short).

1. Initialize the list $2, 3, \ldots, n$.
2. For each current smallest unprocessed p (starting at $p = 2$), remove all multiples $2p, 3p, 4p, \ldots$ up to n.
3. Stop once $p^2 > n$ (i.e. $p > \lfloor \sqrt{n} \rfloor$). Remaining numbers are primes.

Note. 1 is neither prime nor composite.

	2	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

Primes The primes ≤ 100 are:

2, **3**, **5**, **7**, **11**, **13**, **17**, **19**, **23**, **29**, **31**, **37**, **41**, **43**, **47**, **53**, **59**, **61**, **67**, **71**, **73**, **79**, **83**, **89**, **97**.

0.1 L1: 1.1 Divisibility (syllabus, expectations)

0.2 L2: 1.2 Prime numbers (open conjectures, breakthroughs)

0.3 L3: 1.3 GCD, 1.4 Euclidean Algorithm

0.4 L4: 1.5 Fundamental Theorem of Arithmetic

0.5 L5: 1.5 Dirichlet's Theorem, 2.1 Congruences

Lecture 5: Dirichlet's Theorem and applications of the FTA

§ Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions) Let $a, b \in \mathbb{Z}$ with $b > 0$ and $\gcd(a, b) = 1$. Then the arithmetic progression

$$a,\ a+b,\ a+2b,\ a+3b,\ \ldots$$

contains infinitely many prime numbers.

§ Remark (on Theorem Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions)) Setting $a = 1$ and $b = 1$ recovers the elementary statement that there are infinitely many primes. (We remark that the full proof of Dirichlet's theorem requires analytic tools and is beyond the scope of this course.)

§ Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4) There are infinitely many primes of the form $4n + 3$, where n is a nonnegative integer.

§ Lemma 1.23 (Product of numbers congruent to 1 modulo 4) Let $a, b \in \mathbb{Z}$. If $a \equiv 1 \pmod 4$ and $b \equiv 1 \pmod 4$, then $ab \equiv 1 \pmod 4$.

Proof. Write $a = 4n_1 + 1$ and $b = 4n_2 + 1$ with $n_1, n_2 \in \mathbb{Z}$. Then

$$ab = (4n_1 + 1)(4n_2 + 1) = 16n_1n_2 + 4n_1 + 4n_2 + 1 = 4(4n_1n_2 + n_1 + n_2) + 1,$$

so $ab \equiv 1 \pmod 4$. This proves the lemma. □

Proof of Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4). Suppose, for contradiction, that there are only finitely many primes congruent to 3 modulo 4. List them as

$$P_1 = 3,\ P_2,\ P_3,\ \ldots,\ P_r,$$

all distinct and each $P_i \equiv 3 \pmod 4$.

Consider the integer

$$N := 4P_1P_2 \cdots P_r - 1.$$

Note that $N \equiv -1 \equiv 3 \pmod 4$.

Let q be any prime divisor of N. We make two observations:

1. For each $i = 1, \ldots, r$ we have $N \equiv -1 \pmod{P_i}$, so $P_i \nmid N$. Thus q is not equal to any of the P_i.
2. Since $N \equiv 3 \pmod 4$, it is not possible that every prime divisor of N is congruent to 1 modulo 4. Indeed, a product of integers each congruent to $1 \pmod 4$ is itself congruent to $1 \pmod 4$ by Lemma Lemma 1.23 (Product of numbers congruent to 1 modulo 4), contradicting $N \equiv 3 \pmod 4$. Hence some prime divisor q of N satisfies $q \equiv 3 \pmod 4$.

Combining the two observations, q is a prime congruent to 3 modulo 4 that is not among $P_1, \ldots, P_r$, contradicting the assumed finiteness of such primes. Therefore there are infinitely many primes congruent to 3 modulo 4. □

§ Remark (on Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4)) The method above—building an integer with residue $3 \pmod 4$ whose prime divisors cannot all be from a given finite list—gives a proof in this special case of Dirichlet's theorem. However, this argument does not extend directly to all arithmetic progressions; for the full theorem one needs deeper tools (Dirichlet characters and L-series).

Congruences

§ Historical remark (Gauss and congruences) Up to the beginning of the nineteenth century number theory consisted of many isolated results. In 1801, Gauss, in his *Disquisitiones Arithmeticae*, introduced the modern theory of congruences and unified many isolated results in number theory.

§ Definition (Congruence modulo m) Let $a, b, m \in \mathbb{Z}$ with $m > 0$. We say that *a is congruent to b modulo m*, and write

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$. The integer m is called the *modulus* of the congruence.

§ Remark (notation) The notation $a \not\equiv b \pmod{m}$ means $m \nmid (a - b)$. The modulus is customarily taken to be positive; we will assume $m > 0$ in the sequel.

§ Proposition 2.1 (Congruence modulo m is an equivalence relation) Congruence modulo m is reflexive, symmetric, and transitive on $\mathbb{Z}$; hence it is an equivalence relation.

Proof. Reflexive: For any $a \in \mathbb{Z}$, $a - a = 0$ and $m \mid 0$, so $a \equiv a \pmod{m}$.

Symmetric: If $a \equiv b \pmod{m}$ then $m \mid (a - b)$, hence $m \mid -(a - b) = (b - a)$, so $b \equiv a \pmod{m}$.

Transitive: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $m \mid (a - b)$ and $m \mid (b - c)$, therefore $m \mid (a - b) + (b - c) = a - c$, so $a \equiv c \pmod{m}$. $\square$

§ Consequence 2.2 (Partition into residue classes) The set $\mathbb{Z}$ is partitioned into equivalence classes under congruence modulo m.

§ Definition (Notation for residue class) For $x \in \mathbb{Z}$, write $[x]_m$ (or simply $[x]$ when m is understood) for the equivalence class of x modulo m:

$$[x]_m = \{y \in \mathbb{Z} : y \equiv x \pmod{m}\}.$$

(Do not confuse $[x]_m$ with the floor function.)

Example: The equivalence classes modulo 4

$$\begin{aligned}
[0]_4 &= \{x \in \mathbb{Z} : x \equiv 0 \pmod{4}\} = \{4n : n \in \mathbb{Z}\} = \{\dots, -8, -4, 0, 4, 8, \dots\},\\
[1]_4 &= \{x : x = 4n + 1,\ n \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\},\\
[2]_4 &= \{x : x = 4n + 2,\ n \in \mathbb{Z}\} = \{\dots, -6, -2, 2, 6, 10, \dots\},\\
[3]_4 &= \{x : x = 4n + 3,\ n \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}.
\end{aligned}$$

Thus $\mathbb{Z}$ is partitioned into four residue classes modulo 4, and every integer is congruent to exactly one of $0, 1, 2, 3$ modulo 4.

§ Definition (Complete residue system modulo m) A set $R \subset \mathbb{Z}$ is a *complete residue system modulo m* if every integer is congruent modulo m to exactly one element of R.

§ Example (complete residue systems) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m. Also, for example, $\{6, -11, 19, 1988\}$ is a complete residue system modulo 4, since $6 \equiv 2$, $-11 \equiv 1$, $19 \equiv 3$, and $1988 \equiv 0 \pmod{4}$.

§ Proposition 2.3 (Standard complete residue system) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m.

Proof. Let $a \in \mathbb{Z}$. By the Division Algorithm there exist unique $q, r \in \mathbb{Z}$ with $0 \leq r < m$ such that $a = mq + r$. Then $a - r = mq$, so $m \mid (a - r)$ and $a \equiv r \pmod{m}$, with $r \in \{0, 1, \ldots, m-1\}$. This shows every integer is congruent to at least one element of $\{0, 1, \ldots, m-1\}$.

To show uniqueness: suppose $r_1, r_2 \in \{0, 1, \ldots, m-1\}$ and $r_1 \equiv r_2 \pmod{m}$. Then $m \mid (r_1 - r_2)$, but $-(m-1) \leq r_1 - r_2 \leq m - 1$. The only multiple of m in this range is 0, so $r_1 - r_2 = 0$ and hence $r_1 = r_2$. Thus each integer is congruent to exactly one element of $\{0, 1, \ldots, m-1\}$. $\square$

§ Proposition 2.4 (Arithmetic with congruences) Let $a, b, c, d \in \mathbb{Z}$ and $m > 0$. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$,
2. $ac \equiv bd \pmod{m}$.

Proof. From $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ we have $m \mid (a - b)$ and $m \mid (c - d)$.

For (1): $m \mid (a - b) + (c - d) = (a + c) - (b + d)$, hence $a + c \equiv b + d \pmod{m}$.

For (2): Note $m \mid (a - b)$ implies $m \mid c(a - b)$, and $m \mid (c - d)$ implies $m \mid b(c - d)$. Add these divisibilities to get $m \mid c(a - b) + b(c - d) = ac - bd$, so $ac \equiv bd \pmod{m}$. $\square$

0.6 L6: 2.1 Congruences, 2.2 Linear congruences (1 var.)

L6: Congruences

§ Remark (Proposition Proposition 2.4 (Arithmetic with congruences) in class terms) Recall that Proposition 2.4 (Arithmetic with congruences) from the previous lecture can be expressed in terms of equivalence classes modulo m: for residue classes $[a], [c] \in \mathbb{Z}_m$,

$$[a] + [c] = [a + c] \qquad \text{and} \qquad [a][c] = [ac].$$

This viewpoint allows construction of addition and multiplication tables for residue classes (modulo m) using any complete residue system.

Examples: tables for $\mathbb{Z}_4$ and $\mathbb{Z}_5$

$+$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[1]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$[0]_4$
$[2]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[1]_4$
$[3]_4$	$[3]_4$	$[0]_4$	$[1]_4$	$[2]_4$

$\cdot$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$
$[1]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[2]_4$	$[0]_4$	$[2]_4$	$[0]_4$	$[2]_4$
$[3]_4$	$[0]_4$	$[3]_4$	$[2]_4$	$[1]_4$

§ Remark (structure of $\mathbb{Z}_4$) Set $\mathbb{Z}_4 := \{[0]_4, [1]_4, [2]_4, [3]_4\}$. With the above addition and multiplication it is a ring, but not a field: not every nonzero element has a multiplicative inverse (e.g., $[2]_4$ is a zero divisor).

$+$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[1]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$
$[2]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$
$[3]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$
$[4]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$

$\cdot$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$
$[1]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[2]_5$	$[0]_5$	$[2]_5$	$[4]_5$	$[1]_5$	$[3]_5$
$[3]_5$	$[0]_5$	$[3]_5$	$[1]_5$	$[4]_5$	$[2]_5$
$[4]_5$	$[0]_5$	$[4]_5$	$[3]_5$	$[2]_5$	$[1]_5$

§ Remark (structure of $\mathbb{Z}_5$) Let $\mathbb{Z}_5 := \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$. Because 5 is prime every nonzero class has a multiplicative inverse, so $\mathbb{Z}_5$ is a field.

§ Question (why extra structure?) Why does $\mathbb{Z}_5$ have more structure (a field) than $\mathbb{Z}_4$ (only a ring)?

§ Answer (primality of modulus) Essentially: $\mathbb{Z}_m$ is a field if and only if m is prime. If m is composite, then some nonzero classes are zero divisors (no inverses). If $m = p$ is prime, then every nonzero class has an inverse (standard proof via Bezout's identity).

Cancellation and a useful proposition

§ Proposition 2.5 (Cancellation in congruences) Let $a, b, c, m \in \mathbb{Z}$ with $m > 0$, and let $d = \gcd(c, m)$. Then

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{m/d}.$$

Proof. Suppose $ca \equiv cb \pmod{m}$, i.e. $m \mid c(a-b)$. Let $d = \gcd(c, m)$. Write $c = d \cdot c'$ and $m = d \cdot m'$ so that $\gcd(c', m') = 1$. From $m \mid c(a-b)$ we get $dm' \mid dc'(a-b)$, hence $m' \mid c'(a-b)$. Since $\gcd(c', m') = 1$, it follows that $m' \mid (a-b)$, i.e. $a \equiv b \pmod{m'}$, which is $a \equiv b \pmod{m/d}$.

Conversely, if $a \equiv b \pmod{m/d}$ then $m/d \mid (a-b)$, so $m/d \cdot d = m \mid d(a-b)$. But $d \mid c$, so $c(a-b)$ is a multiple of $d(a-b)$, hence $m \mid c(a-b)$ and $ca \equiv cb \pmod{m}$. This proves the equivalence. $\square$

Linear congruences in one variable

§ Definition (linear congruence) A congruence of the form

$$ax \equiv b \pmod{m},$$

with unknown $x \in \mathbb{Z}$ and given integers a, b, m with $m > 0$, is called a *linear congruence* in the variable x.

§ Examples (small checks)

- $2x \equiv 3 \pmod 4$ has no solution: checking $x \equiv 0, 1, 2, 3 \pmod 4$ yields no solution.
- $2x \equiv 3 \pmod 5$ has the unique solution $x \equiv 4 \pmod 5$.
- $2x \equiv 4 \pmod 6$ has two incongruent solutions modulo 6: $x \equiv 2, 5 \pmod 6$.
- $3x \equiv 9 \pmod 6$ has three incongruent solutions modulo 6: $x \equiv 1, 3, 5 \pmod 6$.

§ Theorem 2.6 (Solvability and number of solutions of linear congruences) Let $a, b, m \in \mathbb{Z}$ with $m > 0$ and let $d = \gcd(a, m)$. Then:

1. The congruence $ax \equiv b \pmod m$ has a solution iff $d \mid b$.
2. If $d \mid b$ then there are exactly d incongruent solutions modulo m. Concretely, if x_0 is one solution, then all solutions are
$$x = x_0 + \frac{m}{d} n, \qquad n \in \mathbb{Z},$$
and the classes $x_0 + \frac{m}{d} n$ for $n = 0, 1, \dots, d-1$ give a complete set of incongruent solutions modulo m.

Proof. We use the standard equivalence between the congruence and a linear Diophantine equation. The congruence $ax \equiv b \pmod m$ is equivalent to the existence of an integer y such that

$$ax - my = b. \tag{*}$$

Thus solvability of the congruence is equivalent to solvability of the linear Diophantine equation $(*)$.

(1) If $(*)$ has an integer solution, then $d = \gcd(a, m)$ divides the left-hand side $ax - my$ and hence divides b. Conversely, if $d \mid b$, write $b = de$ for some $e \in \mathbb{Z}$. By Bezout's identity (see **??** or the standard result), there exist integers r, s with $ar + ms = d$. Multiply both sides by e to obtain $are + mse = de = b$. Then $x_0 = re$ and $y_0 = -se$ solve $(*)$, so a solution exists.

(2) Fix any solution x_0 of $ax \equiv b \pmod m$. For any integer n consider

$$x = x_0 + \frac{m}{d} n.$$

Then

$$ax = ax_0 + a\frac{m}{d} n = ax_0 + \frac{a}{d} mn \equiv ax_0 \equiv b \pmod{m},$$

so each such x is a solution. Thus there are infinitely many integer solutions, all congruent modulo $\frac{m}{d}$ shifts.

Now suppose x is any solution. Then subtracting the equation for x_0 from the equation for x yields

$$a(x - x_0) \equiv 0 \pmod{m},$$

i.e. $m \mid a(x - x_0)$. Write $a = da'$ and $m = dm'$, with $\gcd(a', m') = 1$. Then $m'd \mid da'(x - x_0)$, so $m' \mid a'(x - x_0)$. Since $\gcd(a', m') = 1$ we deduce $m' \mid (x - x_0)$, hence $x - x_0 = m'n = \frac{m}{d} n$ for some $n \in \mathbb{Z}$. This shows every solution has the stated form. Two such solutions

$$x_0 + \frac{m}{d} n_1 \quad \text{and} \quad x_0 + \frac{m}{d} n_2$$

are congruent modulo m iff

$$\frac{m}{d}(n_1 - n_2) \equiv 0 \pmod{m} \iff m \mid \frac{m}{d}(n_1 - n_2).$$

Dividing both sides by m/d yields $d \mid (n_1 - n_2)$, so $n_1 \equiv n_2 \pmod{d}$. Therefore choosing $n = 0, 1, \ldots, d-1$ produces exactly d incongruent solutions modulo m. □

§ Corollary 2.7 (Explicit solution set) Under the hypotheses of Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences), if $d = \gcd(a, m)$ and $d \mid b$, then the d incongruent solutions modulo m are

$$x_0 + \frac{m}{d} n, \qquad n = 0, 1, \ldots, d-1,$$

where x_0 is any particular solution.

Worked example

§ Example (solve $16x \equiv 8 \pmod{28}$) We solve $16x \equiv 8 \pmod{28}$. Compute $d = \gcd(16, 28)$.

Euclidean algorithm:

$$28 = 1 \cdot 16 + 12, \qquad 16 = 1 \cdot 12 + 4, \qquad 12 = 3 \cdot 4 + 0,$$

so $d = 4$. Since $4 \mid 8$, the congruence has $d = 4$ incongruent solutions modulo 28 by Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences).

One method: divide the congruence by $d = 4$ to get (working modulo $28/4 = 7$)

$$4x \equiv 2 \pmod{7}.$$

Since $4^{-1} \equiv 2 \pmod{7}$ (because $4 \cdot 2 = 8 \equiv 1 \pmod{7}$), multiply both sides by 2:

$$x \equiv 4 \pmod{7}.$$

Thus the solutions modulo 28 are

$$x \equiv 4 + 7n, \qquad n = 0, 1, 2, 3,$$

which gives the incongruent solutions $x \equiv 4, 11, 18, 25 \pmod{28}$.

(Alternatively, one can find an explicit x_0 via the extended Euclidean computation: from $4 = 2 \cdot 16 - 1 \cdot 28$ we get $8 = 4 \cdot 16 - 2 \cdot 28$, so $x_0 = 4$ is a particular solution, yielding the same solution set.)

Multiplicative inverses

§ Definition (multiplicative inverse modulo m) If $a \in \mathbb{Z}$ and there exists $x \in \mathbb{Z}$ with $ax \equiv 1 \pmod{m}$, then x is called a *multiplicative inverse* of a modulo m (often denoted $a^{-1} \pmod{m}$).

§ Corollary 2.8 (When inverses exist) The congruence $ax \equiv 1 \pmod{m}$ has a solution iff $\gcd(a, m) = 1$. In that case the inverse is unique modulo m.

Proof. Apply Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences) with $b = 1$. If $d = \gcd(a, m) > 1$ then $d \nmid 1$ so there is no solution. If $d = 1$ then there is exactly one solution modulo m. □

§ Example (solve $5x \equiv 12 \pmod{16}$) Since $\gcd(5, 16) = 1$, the congruence has a unique solution modulo 16 by Corollary Corollary 2.8 (When inverses exist). One checks that $5^{-1} \pmod{16} = 13$ because $5 \cdot 13 = 65 \equiv 1 \pmod{16}$. Multiplying by 13 gives

$$x \equiv 13 \cdot 12 \pmod{16}.$$

Compute $13 \cdot 12 = 156 \equiv 12 \pmod{16}$ (since $156 - 144 = 12$), so $x \equiv 12 \pmod{16}$.

0.7 L7: 2.3 Chinese Remainder Thm, 2.4 Wilson's Thm, 2.5 Fermat's Little Thm

L7: The Chinese Remainder Theorem, Wilson's Theorem, Fermat's Little Theorem

Example (motivation). Find a positive integer x with

$$x \equiv 2 \pmod 3, \qquad x \equiv 1 \pmod 4, \qquad x \equiv 3 \pmod 5.$$

This is a system of linear congruences in one variable.

§ Theorem 2.9 (Chinese Remainder Theorem) Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers, and let $b_1, \dots, b_n$ be arbitrary integers. Then the system

$$x \equiv b_i \pmod{m_i} \qquad (i = 1, \dots, n)$$

has a solution, and any two solutions are congruent modulo $M := m_1 m_2 \cdots m_n$. Hence the solution is unique modulo M.

Proof. Let $M := \prod_{i=1}^{n} m_i$ and define $M_i := M/m_i$. Since $(M_i, m_i) = 1$ by the pairwise coprimality hypothesis, by Corollary 2.8 (When inverses exist) (existence of inverses when gcd is 1) there exists x_i with

$$M_i x_i \equiv 1 \pmod{m_i}.$$

Put

$$x := \sum_{i=1}^{n} b_i M_i x_i.$$

For a fixed index i, reduce modulo m_i: every term $b_j M_j x_j$ with $j \neq i$ is a multiple of m_i (since $m_i \mid M_j$ for $j \neq i$), while $M_i x_i \equiv 1 \pmod{m_i}$ by construction. Thus $x \equiv b_i \pmod{m_i}$ for each i, so x is a solution.

If x' is another solution, then $m_i \mid (x - x')$ for every i, and since the m_i are pairwise coprime it follows that $M \mid (x - x')$. Hence $x \equiv x' \pmod M$, proving uniqueness modulo M. □

Example (continued). Here $m_1 = 3, m_2 = 4, m_3 = 5$, so $M = 60$ and

$$M_1 = 60/3 = 20, \qquad M_2 = 60/4 = 15, \qquad M_3 = 60/5 = 12.$$

Solve $M_i x_i \equiv 1 \pmod{m_i}$:

$$20x_1 \equiv 1 \pmod 3 \implies 2x_1 \equiv 1 \pmod 3 \implies x_1 \equiv 2 \pmod 3,$$

so choose $x_1 = 2$.

$$15x_2 \equiv 1 \pmod 4 \implies 3x_2 \equiv 1 \pmod 4 \implies x_2 \equiv 3 \pmod 4,$$

so choose $x_2 = 3$.

$$12x_3 \equiv 1 \pmod 5 \implies 2x_3 \equiv 1 \pmod 5 \implies x_3 \equiv 3 \pmod 5,$$

so choose $x_3 = 3$.

Now

$$x = b_1 M_1 x_1 + b_2 M_2 x_2 + b_3 M_3 x_3 = 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 3 \cdot 12 \cdot 3 = 80 + 45 + 108 = 233.$$

Hence $x \equiv 233 \equiv 53 \pmod{60}$, and the least positive solution is 53.

§ Lemma 2.10 (Self-inverse modulo a prime) Let p be a prime and $a \in \mathbb{Z}$. Then a is its own inverse modulo p (i.e. $a^2 \equiv 1 \pmod{p}$) if and only if $a \equiv \pm 1 \pmod{p}$.

Proof. If $a^2 \equiv 1 \pmod{p}$ then $p \mid (a-1)(a+1)$. Since p is prime, $p \mid (a-1)$ or $p \mid (a+1)$, so $a \equiv 1$ or $a \equiv -1 \pmod{p}$. The converse is immediate. □

§ Theorem 2.11 (Wilson's Theorem) Let p be a prime. Then

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. The result is trivial for $p = 2, 3$, so assume $p \geq 5$. The integers $1, 2, \ldots, p-1$ can be paired into multiplicative inverses modulo p. By Lemma 0.7, the only elements that are their own inverses are 1 and $p-1$. Thus, aside from 1 and $p-1$, the remaining $p-3$ elements pair off, and each pair multiplies to 1 modulo p. Therefore

$$(p-1)! \equiv 1 \cdot (\text{product of inverse-pairs}) \cdot (p-1) \equiv 1 \cdot 1 \cdots 1 \cdot (-1) \equiv -1 \pmod{p},$$

as required. □

§ Remark (converse) The converse is also true: if $n > 1$ and $(n-1)! \equiv -1 \pmod{n}$ then n must be prime.

§ Proposition 2.12 (Converse to Wilson's Theorem) Let $n > 1$ be an integer. If $(n-1)! \equiv -1 \pmod{n}$ then n is prime.

Proof. Suppose n is composite, so $n = ab$ with $1 < a \leq b < n$. Then $a \mid (n-1)!$ because $(n-1)!$ contains the factor a, and hence $(n-1)! \equiv 0 \pmod{a}$. The hypothesis $(n-1)! \equiv -1 \pmod{n}$ implies $(n-1)! \equiv -1 \pmod{a}$ (since $a \mid n$), so $-1 \equiv 0 \pmod{a}$, which is impossible. Therefore no such nontrivial factor a exists and n must be prime. □

§ Remark (Wilson primes) A prime p is called a *Wilson prime* if $(p-1)! \equiv -1 \pmod{p^2}$. Only three Wilson primes are known: $5, 13, 563$. It is unknown whether infinitely many (or only finitely many) Wilson primes exist.

§ Theorem 2.13 (Fermat's Little Theorem) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the $p-1$ integers $a, 2a, \ldots, (p-1)a$. None of these is congruent to $0 \pmod{p}$ (since $p \nmid a$), and no two are congruent modulo p (if $ia \equiv ja \pmod{p}$ with $1 \leq i < j \leq p-1$, multiply by the inverse of a modulo p, which exists by Corollary 2.8 (When inverses exist), to deduce $i \equiv j \pmod{p}$, impossible). Hence the residues of $a, 2a, \ldots, (p-1)a$ modulo p are a permutation of $1, 2, \ldots, p-1$. Multiplying all these congruences yields

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since $p \nmid (p-1)!$ (true for prime p), we may cancel $(p-1)!$ from both sides to obtain $a^{p-1} \equiv 1 \pmod{p}$. (Note: this cancellation step uses only that $\gcd((p-1)!, p) = 1$; Wilson's Theorem is not required.) □

0.8 L8: 2.5 Pseudoprimes, 2.6 Euler's Thm, 3.1 Arithmetic functions/multiplicativity

L8: Corollaries to Fermat's Little Theorem, Pseudoprimes, Euler's Theorem, Arithmetic Functions

§ Corollary 2.14 (Inverse modulo a prime) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then a^{p-2} is the multiplicative inverse of a modulo p; that is,

$$a \cdot a^{p-2} \equiv 1 \pmod{p}.$$

Proof. By Fermat's Little Theorem (Theorem 0.7), $a^{p-1} \equiv 1 \pmod{p}$. Multiply both sides by a^{-1} if you like, or simply rewrite $a^{p-1} = a \cdot a^{p-2}$ to get $a \cdot a^{p-2} \equiv 1 \pmod{p}$. Thus a^{p-2} is an inverse of a modulo p. □

§ Remark (application of Corollary Corollary 2.14 (Inverse modulo a prime)) Corollary Corollary 2.14 (Inverse modulo a prime) is useful for solving linear congruences $ax \equiv b \pmod{p}$ when p is prime and $p \nmid a$: multiply both sides by a^{p-2} to obtain the solution.

§ Corollary 2.15 (Fresh form of Fermat) Let p be prime and $a \in \mathbb{Z}$. Then

$$a^p \equiv a \pmod{p}.$$

Proof. If $p \mid a$ the congruence is trivial. If $p \nmid a$, multiply Fermat's Little Theorem $a^{p-1} \equiv 1 \pmod{p}$ by a to get $a^p \equiv a \pmod{p}$. □

§ Corollary 2.16 (case $a = 2$) For every prime p,

$$2^p \equiv 2 \pmod{p}.$$

Proof. Take $a = 2$ in Corollary Corollary 2.15 (Fresh form of Fermat). □

§ Question (converse for base 2) Is the converse of Corollary Corollary 2.16 (case $a = 2$) true? That is, if $n > 1$ and $2^n \equiv 2 \pmod{n}$, must n be prime?

§ Answer and counterexample (341) No. Example: $n = 341 = 11 \cdot 31$ is composite but satisfies $2^{341} \equiv 2 \pmod{341}$. Since $(11, 31) = 1$, it suffices to check modulo 11 and 31.

By Fermat's Little Theorem,

$$2^{10} \equiv 1 \pmod{11} \quad \Rightarrow \quad 2^{341} \equiv (2^{10})^{34} \cdot 2 \equiv 1^{34} \cdot 2 \equiv 2 \pmod{11},$$

and

$$2^{30} \equiv 1 \pmod{31} \quad \Rightarrow \quad 2^{341} \equiv (2^{30})^{11} \cdot 2^{11} \equiv 1^{11} \cdot 2^{11} \pmod{31}.$$

Since $2^5 \equiv -1 \pmod{31}$ (or compute directly), we have $2^{11} \equiv 2 \pmod{31}$, so $2^{341} \equiv 2 \pmod{31}$. Hence $2^{341} \equiv 2 \pmod{341}$ although 341 is composite.

§ Definition (pseudoprime to base 2) A composite integer $n > 1$ is called a *(base-2) pseudoprime* if

$$2^n \equiv 2 \pmod{n}.$$

§ Remark (pseudoprimes) 341 is the smallest base-2 pseudoprime. There are infinitely many pseudoprimes and they arise for different bases; pseudoprimes show that the naive converse of Fermat's Little Theorem is false.

§ Definition (Euler's phi-function) For $n \in \mathbb{Z}_{>0}$ the Euler totient function $\phi(n)$ is the number of integers $1 \le x \le n$ with $\gcd(x, n) = 1$; equivalently,

$$\phi(n) := \#\{x \in \{1, 2, \ldots, n\} : \gcd(x, n) = 1\}.$$

§ Examples (values of ϕ)

$$\phi(12) = 4 \quad (\text{the classes } 1, 5, 7, 11), \qquad \phi(14) = 6 \quad (\text{the classes } 1, 3, 5, 9, 11, 13),$$

and for a prime p, $\phi(p) = p - 1$ (classes $1, 2, \ldots, p - 1$).

§ Theorem 2.17 (Euler's Theorem) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

Proof. Let $r_1, r_2, \ldots, r_{\phi(m)}$ be the positive integers $\le m$ that are relatively prime to m. Since $\gcd(a, m) = 1$, multiplication by a permutes this reduced residue system: the numbers $ar_1, \ldots, ar_{\phi(m)}$ are all distinct modulo m and each is relatively prime to m (see Lemma 1.5 and Lemma 1.14 referenced earlier). Hence the set $\{ar_i \bmod m\}$ is the same set of residues as $\{r_i\}$. Multiplying all residues yields

$$a^{\phi(m)} \prod_{i=1}^{\phi(m)} r_i \equiv \prod_{i=1}^{\phi(m)} r_i \pmod{m}.$$

Since $\gcd\left(\prod_i r_i,\, m\right) = 1$, we may cancel the product to obtain $a^{\phi(m)} \equiv 1 \pmod{m}$. □

§ Remark (special case) Fermat's Little Theorem is the special case $m = p$ prime, since $\phi(p) = p - 1$.

§ Definition (reduced residue system) A set of $\phi(m)$ integers each relatively prime to m and pairwise incongruent modulo m is called a *reduced residue system* modulo m.

§ Example (reduced residue systems) $\{1, 5, 7, 11\}$ is a reduced residue system modulo 12. If a is coprime to m, then $\{ar_1, \ldots, ar_{\phi(m)}\}$ is also a reduced residue system modulo m (this was the idea used in the proof of Euler's Theorem).

§ Corollary 2.19 (inverse via Euler) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then $a^{\phi(m)-1}$ is the multiplicative inverse of a modulo m.

Proof. From $a^{\phi(m)} \equiv 1 \pmod{m}$ multiply by a^{-1} (or rewrite as $a \cdot a^{\phi(m)-1} \equiv 1$) to see $a^{\phi(m)-1}$ is an inverse of a modulo m. □

§ Remark (use of Corollary Corollary 2.19 (inverse via Euler)) Corollary Corollary 2.19 (inverse via Euler) helps solve linear congruences $ax \equiv b \pmod{m}$ when $\gcd(a, m) = 1$: multiply both sides by $a^{\phi(m)-1}$ to obtain the solution.

§ Definition (arithmetic function) An *arithmetic function* is a function $f : \mathbb{Z}_{>0} \to \mathbb{C}$ (or $\mathbb{R}$, or $\mathbb{Z}$). Examples: the Euler ϕ-function, the divisor-counting function $d(n)$, the sum-of-divisors function $\sigma(n)$, etc.

§ Definition (multiplicative vs completely multiplicative) An arithmetic function f is *multiplicative* if $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is *completely multiplicative* if $f(mn) = f(m)f(n)$ for all positive integers m, n (no coprimality required).

§ Proposition (values determined by prime powers) If f is multiplicative and

$$n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$$

is the prime factorization of n, then

$$f(n) = \prod_{i=1}^{r} f\big(p_i^{a_i}\big).$$

If f is completely multiplicative, then

$$f(n) = \prod_{i=1}^{r} f(p_i)^{a_i}.$$

Proof. Write n as a product of relatively prime prime-power factors and apply multiplicativity repeatedly. □

§ Remark (implication) Thus knowing a multiplicative arithmetic function on prime powers (or a completely multiplicative function on primes) determines it everywhere.

0.9 L9: 3.1 Arithmetic functions/multiplicativity, 3.2 Euler phi function

L9: Arithmetic Functions and Multiplicativity

§ Summation over divisors: notation By

$$\sum_{d|n} f(d)$$

we mean the sum of $f(d)$ over all *positive* divisors d of n. For example,

$$\sum_{d|12} f(d) = f(1) + f(2) + f(3) + f(4) + f(6) + f(12).$$

§ Multiplicative functions and divisor sums Let f be an arithmetic function and define, for $n \in \mathbb{Z}_{>0}$,

$$F(n) := \sum_{d|n} f(d).$$

If f is multiplicative then F is multiplicative.

Proof. Let $m, n \in \mathbb{Z}_{>0}$ with $\gcd(m, n) = 1$. We must show $F(mn) = F(m)\, F(n)$.

Claim: Every positive divisor d of mn can be written uniquely as $d = d_1 d_2$ with $d_1 \mid m$, $d_2 \mid n$ and $\gcd(d_1, d_2) = 1$.

Proof of claim: Write the prime factorizations

$$m = \prod_{i=1}^{r} p_i^{a_i}, \qquad n = \prod_{j=1}^{s} q_j^{b_j},$$

where the primes p_i and q_j are distinct because $\gcd(m, n) = 1$. Then any divisor d of mn has the form

$$d = \prod_{i=1}^{r} p_i^{e_i} \prod_{j=1}^{s} q_j^{f_j},$$

with $0 \le e_i \le a_i$ and $0 \le f_j \le b_j$. Set

$$d_1 = \prod_{i=1}^{r} p_i^{e_i}, \qquad d_2 = \prod_{j=1}^{s} q_j^{f_j}.$$

Clearly $d_1 \mid m$, $d_2 \mid n$, $\gcd(d_1, d_2) = 1$, and $d = d_1 d_2$. Uniqueness follows from uniqueness of prime factorization. □

Using the claim,

$$F(mn) = \sum_{d|mn} f(d) = \sum_{\substack{d_1|m \\ d_2|n}} f(d_1 d_2).$$

Since f is multiplicative and $\gcd(d_1, d_2) = 1$ for every pair occurring in the sum, we have $f(d_1 d_2) = f(d_1) f(d_2)$. Thus

$$F(mn) = \sum_{d_1|m} \sum_{d_2|n} f(d_1) f(d_2) = \Big(\sum_{d_1|m} f(d_1)\Big)\Big(\sum_{d_2|n} f(d_2)\Big) = F(m)F(n).$$

This proves that F is multiplicative. □

§ Example: verification for $m = 3$, $n = 4$ Let $m = 3$ and $n = 4$. The positive divisors are

$$\{\, d_1 \mid 3 \,\} = \{1, 3\}, \qquad \{\, d_2 \mid 4 \,\} = \{1, 2, 4\}.$$

All divisors of 12 are the products $d_1 d_2$ with $d_1 \in \{1,3\}$ and $d_2 \in \{1,2,4\}$; hence

$$\sum_{d|12} f(d) = \sum_{d_1|3} \sum_{d_2|4} f(d_1 d_2) = \sum_{d_1|3} \sum_{d_2|4} f(d_1) f(d_2)$$

(using multiplicativity of f), so

$$\sum_{d|12} f(d) = \Big(\sum_{d_1|3} f(d_1)\Big)\Big(\sum_{d_2|4} f(d_2)\Big) = F(3)F(4).$$

§ The Euler φ-function is multiplicative The Euler totient function $\varphi(n)$ (the number of integers in $\{1,2,\ldots,n\}$ relatively prime to n) is multiplicative: if $\gcd(m,n)=1$ then

$$\varphi(mn) = \varphi(m)\,\varphi(n).$$

Proof. List the integers $1,2,\ldots,mn$ in an $m \times n$ array with rows indexed by $i=1,\ldots,m$ and columns by $k=0,\ldots,n-1$ as

$$\begin{array}{ccccc} 1 & m+1 & 2m+1 & \cdots & (n-1)m+1 \\ 2 & m+2 & 2m+2 & \cdots & (n-1)m+2 \\ \vdots & \vdots & \vdots & & \vdots \\ m & m+m & 2m+m & \cdots & (n-1)m+m \end{array}$$

Consider the i-th row

$$i,\ m+i,\ 2m+i,\ \ldots,\ (n-1)m+i.$$

If $\gcd(i,m)>1$ then every entry of that row shares a common factor with m, and hence cannot be relatively prime to mn. Thus only rows with $\gcd(i,m)=1$ can contain integers relatively prime to mn. There are precisely $\varphi(m)$ such rows.

Fix a row with $\gcd(i,m)=1$. The n entries of that row are distinct modulo n: if $jm+i \equiv km+i \pmod{n}$ with $0 \le j,k \le n-1$, then $(j-k)m \equiv 0 \pmod{n}$. Since $\gcd(m,n)=1$, m has a multiplicative inverse modulo n, so $j \equiv k \pmod{n}$, hence $j=k$. Thus the row is a complete residue system modulo n, so exactly $\varphi(n)$ of its entries are relatively prime to n. Because those entries are relatively prime to both m and n, they are relatively prime to mn.

Therefore each of the $\varphi(m)$ admissible rows contains exactly $\varphi(n)$ integers relatively prime to mn. Thus

$$\varphi(mn) = \varphi(m)\varphi(n).$$

$\square$

§ Values of φ at prime powers Let p be prime and $a \in \mathbb{Z}_{>0}$. Then

$$\varphi(p^a) = p^a - p^{a-1} = p^a\Big(1 - \frac{1}{p}\Big).$$

Proof. Among the p^a integers $1,2,\ldots,p^a$, those not coprime to p^a are exactly the multiples of p:

$$p, 2p, 3p, \ldots, p^{a-1}p,$$

and there are p^{a-1} such multiples. Hence

$$\varphi(p^a) = p^a - p^{a-1} = p^a\Big(1 - \frac{1}{p}\Big).$$

$\square$

§ Product formula for $\varphi(n)$ For $n \in \mathbb{Z}_{>0}$,

$$\varphi(n) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big),$$

where the product is taken over the distinct prime divisors p of n.

Proof. If $n = 1$ the result holds (empty product $= 1$). For $n > 1$, write the prime factorization

$$n = \prod_{i=1}^{r} p_i^{a_i}.$$

By Theorems The Euler φ-function is multiplicative and Values of φ at prime powers,

$$\varphi(n) = \prod_{i=1}^{r} \varphi(p_i^{a_i}) = \prod_{i=1}^{r} p_i^{a_i}\Big(1 - \frac{1}{p_i}\Big) = n \prod_{p \mid n}\Big(1 - \frac{1}{p}\Big).$$

□

§ Example: $\varphi(504)$ Since $504 = 2^3 \cdot 3^2 \cdot 7$, we get

$$\varphi(504) = 504\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{7}\Big).$$

Compute:

$$\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{7}\Big) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{6}{7} = \frac{2}{7},$$

so $\varphi(504) = 504 \cdot \frac{2}{7} = 144$.

§ Gauss's identity for φ For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d \mid n} \varphi(d) = n.$$

Proof. For each positive divisor d of n define

$$S_d := \{\, m \in \mathbb{Z} : 1 \le m \le n,\ \gcd(m, n) = d \,\}.$$

Note that $\gcd(m, n) = d$ if and only if $\gcd(m/d, n/d) = 1$, so the number of integers in S_d equals $\varphi(n/d)$. The sets S_d (as d runs over the positive divisors of n) partition the set $\{1, 2, \ldots, n\}$; hence

$$n = \sum_{d \mid n} |S_d| = \sum_{d \mid n} \varphi(n/d).$$

Reindexing the sum by replacing d with n/d shows

$$n = \sum_{d \mid n} \varphi(d),$$

as claimed. □

§ Example: $n = 12$ For $n = 12$ the divisors and corresponding sets are

$$
\begin{aligned}
S_1 &= \{1,5,7,11\}, & |S_1| &= \varphi(12) = 4,\\
S_2 &= \{2,10\}, & |S_2| &= \varphi(6) = 2,\\
S_3 &= \{3,9\}, & |S_3| &= \varphi(4) = 2,\\
S_4 &= \{4,8\}, & |S_4| &= \varphi(3) = 2,\\
S_6 &= \{6\}, & |S_6| &= \varphi(2) = 1,\\
S_{12} &= \{12\}, & |S_{12}| &= \varphi(1) = 1.
\end{aligned}
$$

Summing gives

$$\sum_{d|12} \varphi(d) = 4 + 2 + 2 + 2 + 1 + 1 = 12,$$

confirming Gauss's identity.

0.10 L10: 3.3 Divisor function, 3.4 Sum of divisors, 3.5 Perfect numbers, 3.6 Möbius function

L10: The number of positive divisors and related arithmetic functions

§ Definition *(Number of positive divisors).* Let $n \in \mathbb{Z}_{>0}$. The *number of positive divisors function*, denoted $V(n)$, is

$$V(n) := \left|\{d \in \mathbb{Z} : \ d > 0 \text{ and } d \mid n\}\right|.$$

§ Remark Many authors denote the divisor-counting function by $\tau(n)$ or $d(n)$. Here we keep your notation $V(n)$ for consistency with the provided notes.

§ Theorem 3.6 *(Multiplicativity of V).* The function $V(n)$ is multiplicative: if $\gcd(m, n) = 1$ then

$$V(mn) = V(m)V(n).$$

Proof. Note that

$$V(n) = \sum_{d|n} 1,$$

i.e. the sum over positive divisors of the constant arithmetic function $f(d) \equiv 1$. Since the constant function $f(d) = 1$ is multiplicative (check: $1 \cdot 1 = 1$), Theorem 0.9 implies that the divisor-sum $V(n)$ is multiplicative. □

§ Theorem 3.7 Let p be a prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$V(p^a) = a + 1.$$

Proof. The positive divisors of p^a are $1, p, p^2, \ldots, p^a$; there are $a + 1$ of them. □

§ Theorem 3.8 Let $n = \prod_{i=1}^{r} p_i^{a_i}$ be the prime factorization of n with distinct primes p_i and exponents $a_i \geq 0$. Then

$$V(n) = \prod_{i=1}^{r} (a_i + 1).$$

Proof. From multiplicativity (Theorem 0.10) and the prime-power values (Theorem 0.10) we obtain

$$V(n) = \prod_{i=1}^{r} V(p_i^{a_i}) = \prod_{i=1}^{r} (a_i + 1).$$

□

Example 0.1. For $504 = 2^3 \cdot 3^2 \cdot 7^1$ we have

$$V(504) = (3 + 1)(2 + 1)(1 + 1) = 4 \cdot 3 \cdot 2 = 24.$$

§ Definition *(Sum-of-divisors function).* For $n \in \mathbb{Z}_{>0}$ define

$$\sigma(n) := \sum_{d|n} d,$$

the sum of the positive divisors of n.

§ Theorem 3.9 *(Multiplicativity of σ).* The function $\sigma(n)$ is multiplicative.

Proof. The arithmetic function $f(d) = d$ is multiplicative (if $\gcd(a, b) = 1$ then $(ab) = a \cdot b$ and hence the function $d \mapsto d$ satisfies $f(ab) = f(a)f(b)$ when considered on coprime arguments). Thus by Theorem 0.9, the divisor-sum $\sigma(n) = \sum_{d|n} d$ is multiplicative. ☐

§ Theorem 3.10 Let p be prime and $a \in \mathbb{Z}_{\geq 0}$. Then

$$\sigma(p^a) = 1 + p + \cdots + p^a = \frac{p^{a+1} - 1}{p - 1}.$$

Proof. Sum the geometric series $1 + p + \cdots + p^a$. ☐

§ Theorem 3.11 If $n = \prod_{i=1}^{r} p_i^{a_i}$ then

$$\sigma(n) = \prod_{i=1}^{r} \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

Proof. Immediate from multiplicativity (Theorem 0.10) and the prime-power formula (Theorem 0.10). ☐

Example 0.2. For $504 = 2^3 \cdot 3^2 \cdot 7$,

$$\sigma(504) = \underbrace{\frac{2^4 - 1}{2 - 1}}_{=15} \cdot \underbrace{\frac{3^3 - 1}{3 - 1}}_{=13} \cdot \underbrace{\frac{7^2 - 1}{7 - 1}}_{=8} = 15 \cdot 13 \cdot 8 = 1560.$$

§ Definition *(Perfect number).* An integer $n > 0$ is *perfect* if $\sigma(n) = 2n$. Equivalently, the sum of proper divisors (divisors $< n$) equals n.

Example 0.3. $\sigma(6) = 1 + 2 + 3 + 6 = 12 = 2 \cdot 6$, so 6 is perfect. $\sigma(28) = 1 + 2 + 4 + 7 + 14 + 28 = 56 = 2 \cdot 28$, so 28 is perfect.

§ Theorem 3.12 *(Characterization of even perfect numbers — Euclid and Euler).* An integer $n > 0$ is an even perfect number if and only if

$$n = 2^{p-1}(2^p - 1),$$

where p is prime and $2^p - 1$ is prime (a Mersenne prime).

Proof. We give both directions.

(Euler: if n is even and perfect then it has the stated form). Let n be an even perfect number. Write $n = 2^{k-1}m$ with $k \geq 2$ and m odd. Then by multiplicativity,

$$\sigma(n) = \sigma(2^{k-1})\sigma(m) = (2^k - 1)\sigma(m).$$

Because n is perfect, $\sigma(n) = 2n = 2^k m$. Therefore

$$(2^k - 1)\sigma(m) = 2^k m.$$

Hence $2^k - 1$ divides the right-hand side, and since $\gcd(2^k - 1, 2^k) = 1$ we have $2^k - 1 \mid m$. Write $m = (2^k - 1)t$ for some integer $t \geq 1$. Substituting gives

$$(2^k - 1)\sigma(m) = 2^k(2^k - 1)t \quad \Longrightarrow \quad \sigma(m) = 2^k t.$$

But $\sigma(m)$ is at least $1 + m$ (since it includes 1 and m), so

$$2^k t = \sigma(m) \geq 1 + m = 1 + (2^k - 1)t = 1 + (2^k - 1)t.$$

Rearranging,

$$2^k t - (2^k - 1)t \geq 1 \implies t \geq 1,$$

and this inequality becomes

$$t \geq 1 + \frac{1}{2^k - 1} > 1 \quad \text{if } t > 1,$$

which contradicts the equality of integers unless $t = 1$. (A cleaner integer argument: if $t > 1$ then the displayed inequality would give $2^k t \geq m + 1 + t > m + t = \sigma(m)$, contradiction.) Thus $t = 1$, so $m = 2^k - 1$. Then $\sigma(m) = m + 1 = 2^k$, so m must be prime. Putting $p = k$ yields

$$n = 2^{p-1}(2^p - 1),$$

with $2^p - 1$ prime.

(Euclid: if p and $2^p - 1$ are prime then $n = 2^{p-1}(2^p - 1)$ is perfect). Suppose p is prime and $q := 2^p - 1$ is prime. Then using multiplicativity and the geometric formula,

$$\sigma(n) = \sigma(2^{p-1})\sigma(q) = (2^p - 1)(1 + q) = (2^p - 1)(1 + 2^p - 1) = (2^p - 1)2^p.$$

But $(2^p - 1)2^p = 2 \cdot 2^{p-1}(2^p - 1) = 2n$, so $\sigma(n) = 2n$. Thus n is perfect. □

§ Remark It is unknown whether there are infinitely many Mersenne primes, and hence unknown whether there are infinitely many even perfect numbers. It is also an open problem whether odd perfect numbers exist.

§ Definition *(Möbius function μ).* For $n \in \mathbb{Z}_{>0}$ define

$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & \text{if } p^2 \mid n \text{ for some prime } p, \\ (-1)^r, & \text{if } n = \prod_{i=1}^r p_i \text{ is a product of } r \text{ distinct primes.} \end{cases}$$

Example 0.4. Since $504 = 2^3 \cdot 3^2 \cdot 7$ contains the square 3^2, $\mu(504) = 0$. Since $30 = 2 \cdot 3 \cdot 5$ is the product of three distinct primes, $\mu(30) = (-1)^3 = -1$.

§ Theorem 3.13 The Möbius function $\mu(n)$ is multiplicative.

Proof. Let $\gcd(m, n) = 1$. If either m or n is divisible by the square of a prime, then so is mn, and both $\mu(m)\mu(n)$ and $\mu(mn)$ equal 0. Otherwise both m and n are squarefree, and if $m = \prod_{i=1}^r p_i$ and $n = \prod_{j=1}^s q_j$ with all primes distinct, then $\mu(m) = (-1)^r$, $\mu(n) = (-1)^s$, and $\mu(mn) = (-1)^{r+s} = \mu(m)\mu(n)$. □

§ Proposition 3.14 For $n \in \mathbb{Z}_{>0}$,

$$\sum_{d \mid n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

Proof. Define $F(n) := \sum_{d \mid n} \mu(d)$. Since μ is multiplicative (Theorem 0.10), Theorem 0.9 implies F is multiplicative, so it suffices to evaluate F on prime powers. Clearly $F(1) = 1$. For a prime p and $a \geq 1$,

$$F(p^a) = \sum_{d \mid p^a} \mu(d) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^a) = 1 + (-1) + 0 + \cdots + 0 = 0.$$

By multiplicativity this gives $F(n) = 0$ for every $n > 1$. □

Example 0.5. For $n = 12$,

$$\sum_{d \mid 12} \mu(d) = \mu(1) + \mu(2) + \mu(3) + \mu(4) + \mu(6) + \mu(12) = 1 + (-1) + (-1) + 0 + 1 + 0 = 0.$$